

CoOOH/BiVO₄ Photocatalyst for Efficient Solar Water Splitting

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Introduction

Photoelectrochemical water splitting is a key green-H₂ route, but its efficiency is limited by the sluggish oxygen evolution reaction (OER). Bismuth vanadate (BiVO₄) is a leading visible-light photoanode (band gap ~2.2-2.4 eV) whose slow surface kinetics and fast electron-hole recombination cap its solar-to-H₂ efficiency below the theoretical limit. Interfacing BiVO₄ with a cobalt oxyhydroxide (CoOOH) co-catalyst mitigates these losses: the surface Co³⁺ centres, tunable between low- and intermediate-spin states, provide a direct lever on the e_g descriptor of OER

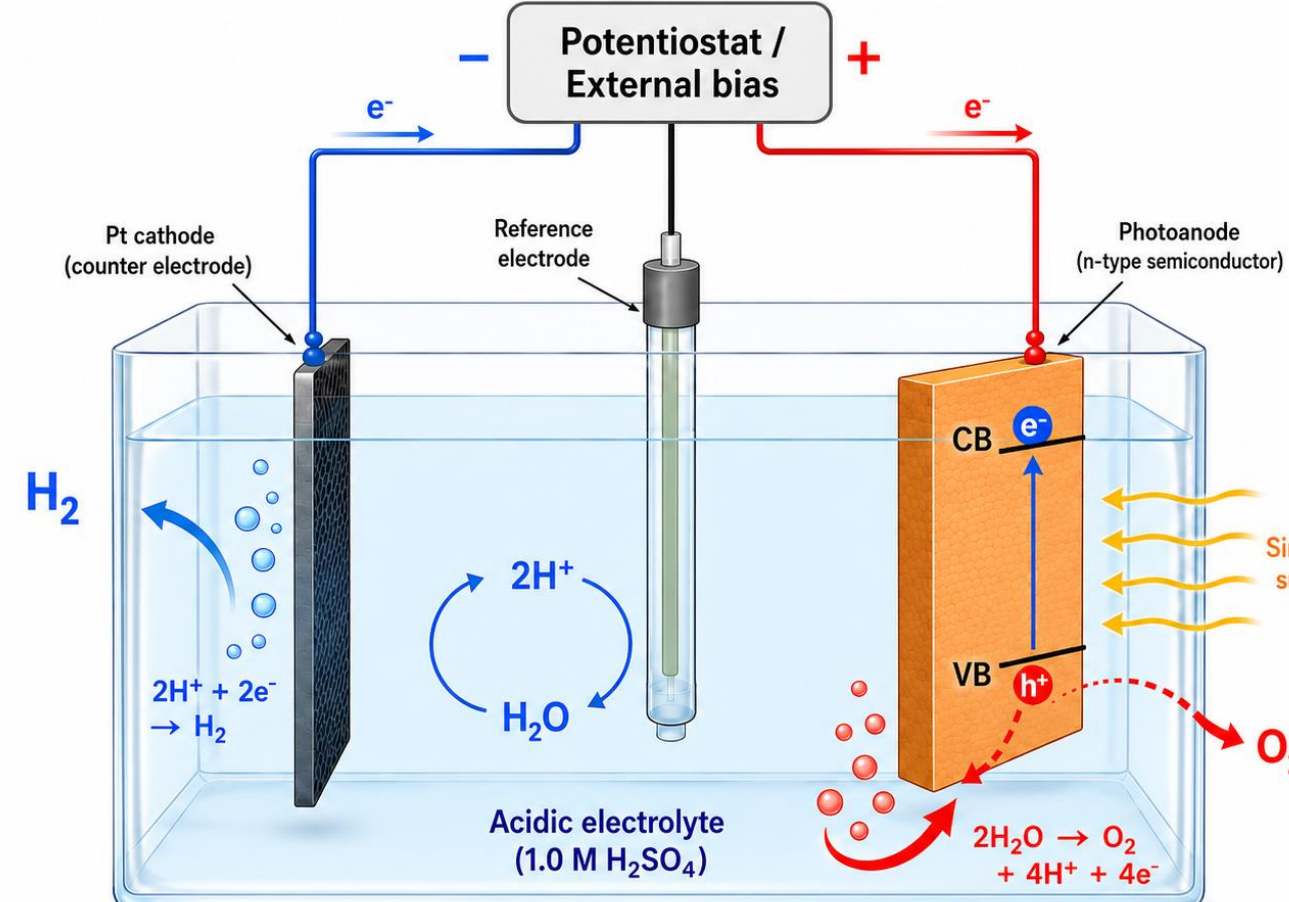


Figure 1. PEC water-splitting cell with a BiVO₄-based photoanode and Pt cathode.

The DFT bottleneck: Meta-GGA r²SCAN calculations become computationally prohibitive for exploring the configurational space of complex CoOOH/BiVO₄ heterostructures. Universal MLIPs, **MACE-MH-1** and **ORB-V3**, offer near-DFT accuracy at a fraction of the computational cost but require system-specific fine-tuning.

Goal: Benchmark fine-tuned **MACE-MH-1** and **ORB-V3** against r²SCAN to evaluate their accuracy, transferability, and scalability for predicting OER energetics, then use the best-performing model to efficiently explore the CoOOH/BiVO₄ configurational space beyond the reach of routine DFT.

Fine-Tuned MLIPs

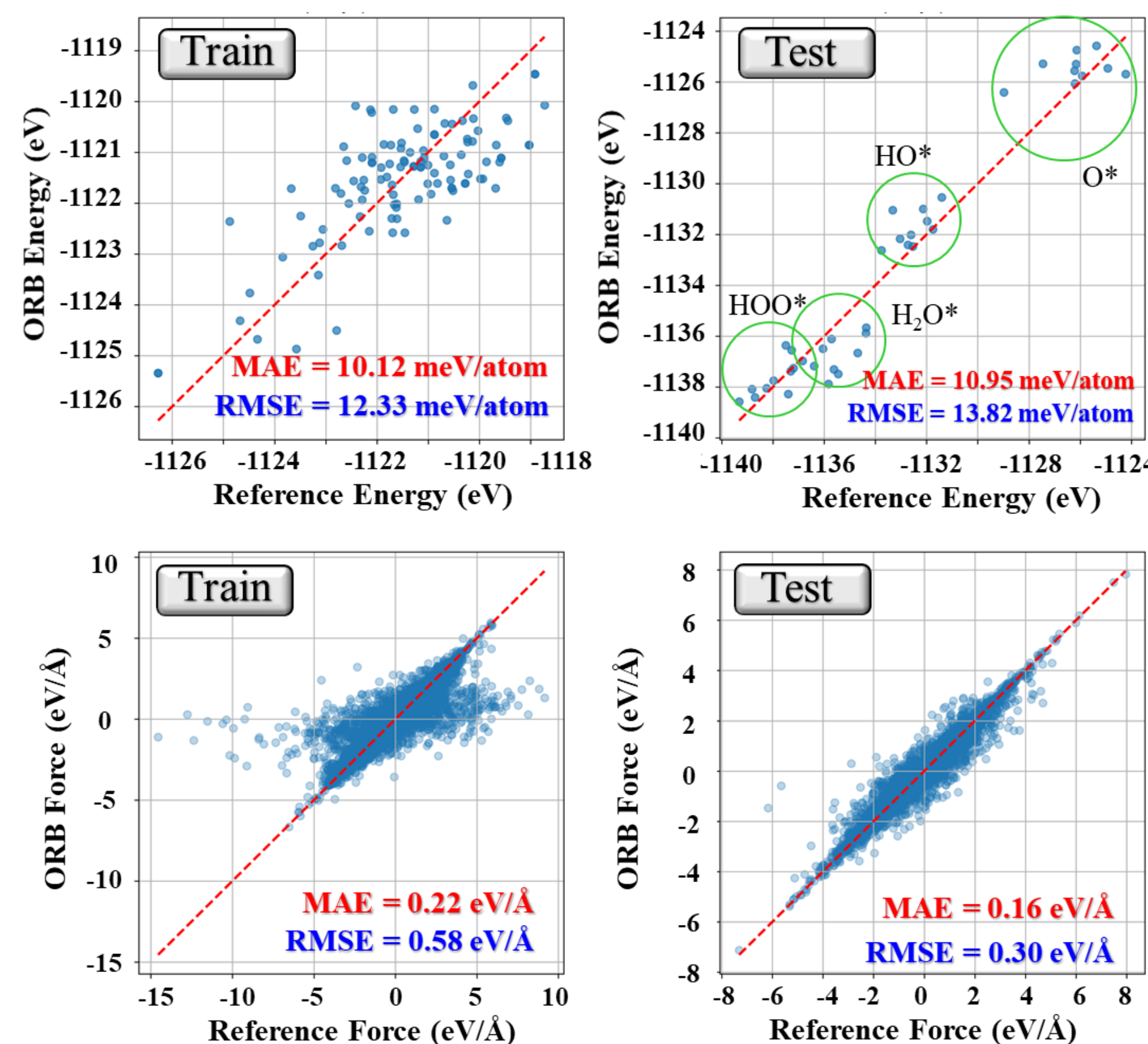


Figure 2. Parity plots for fine-tuned conservative ORB-V3 on CoOOH/BiVO₄. The same fine-tuning protocol was applied to MACE-MH-1.

➤ Excellent transferability to unseen CoOOH/BiVO₄ + OER configurations.

➤ ORB-V3 conservative delivers the lowest energy and force errors.

➤ ORB-V3 reproduces r²SCAN OER free energies within ~0.15 eV for all four PCET steps.

➤ **ORB-V3** → The optimal MLIP for accurate, transferable, and scalable CoOOH/BiVO₄ simulations.

Adsorption Energies: MLIP - DFT

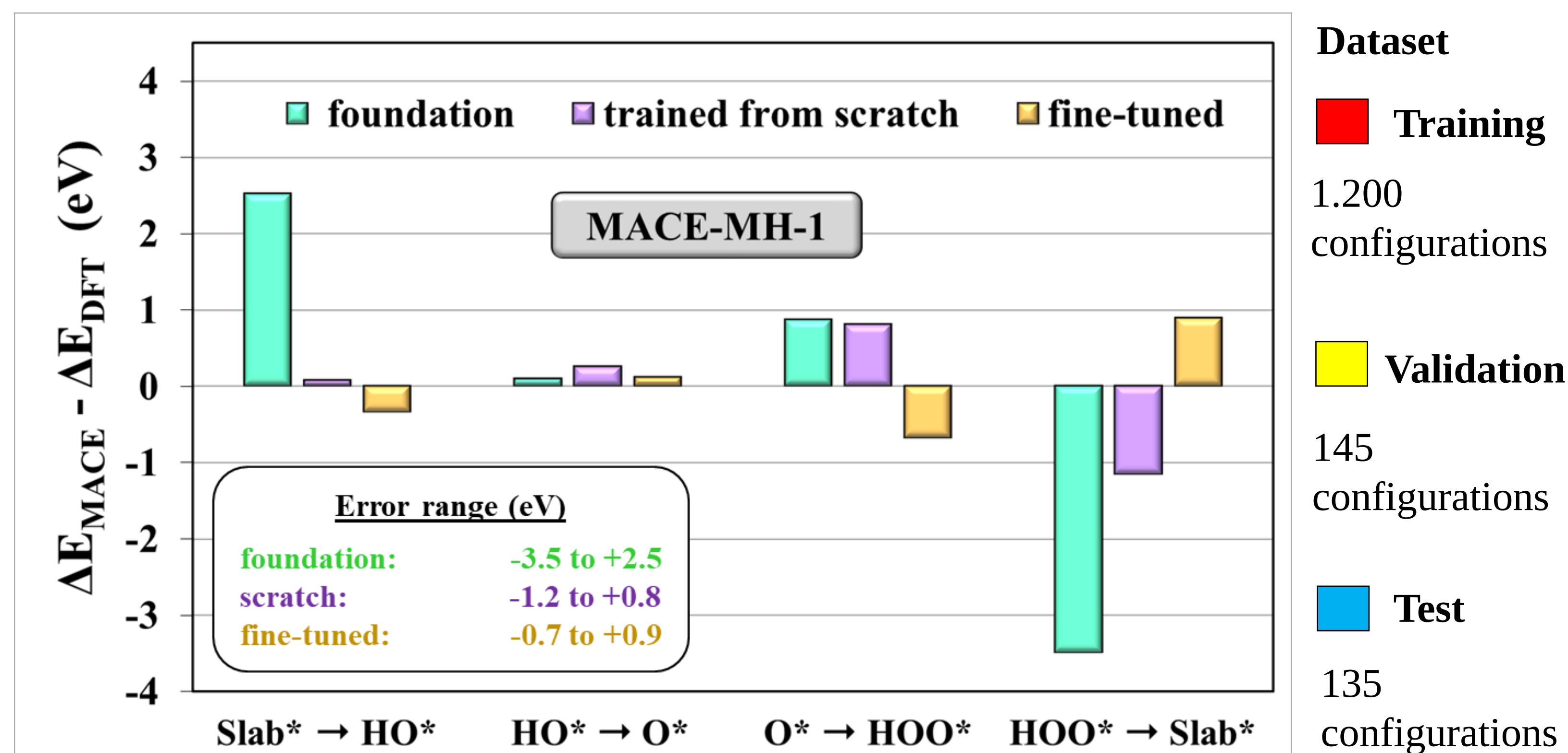


Figure 3. Deviations of MACE-MH-1 adsorption energies from r²SCAN for the four OER steps on CoOOH/BiVO₄.

- **Foundation MACE-MH-1** exhibits large deviations from r²SCAN for our Co-containing systems.
- **Training from scratch** reduces errors but remains insufficient for quantitative OER energetics.
- **Fine-tuned MACE-MH-1** closely reproduces r²SCAN free energies across all OER intermediates for CoOOH/BiVO₄.
- **System-specific fine-tuning** is essential for accurate prediction of interfacial OER energetics and reliable MLIP screening.

Dataset

Training

1,200 configurations

Validation

145 configurations

Test

135 configurations

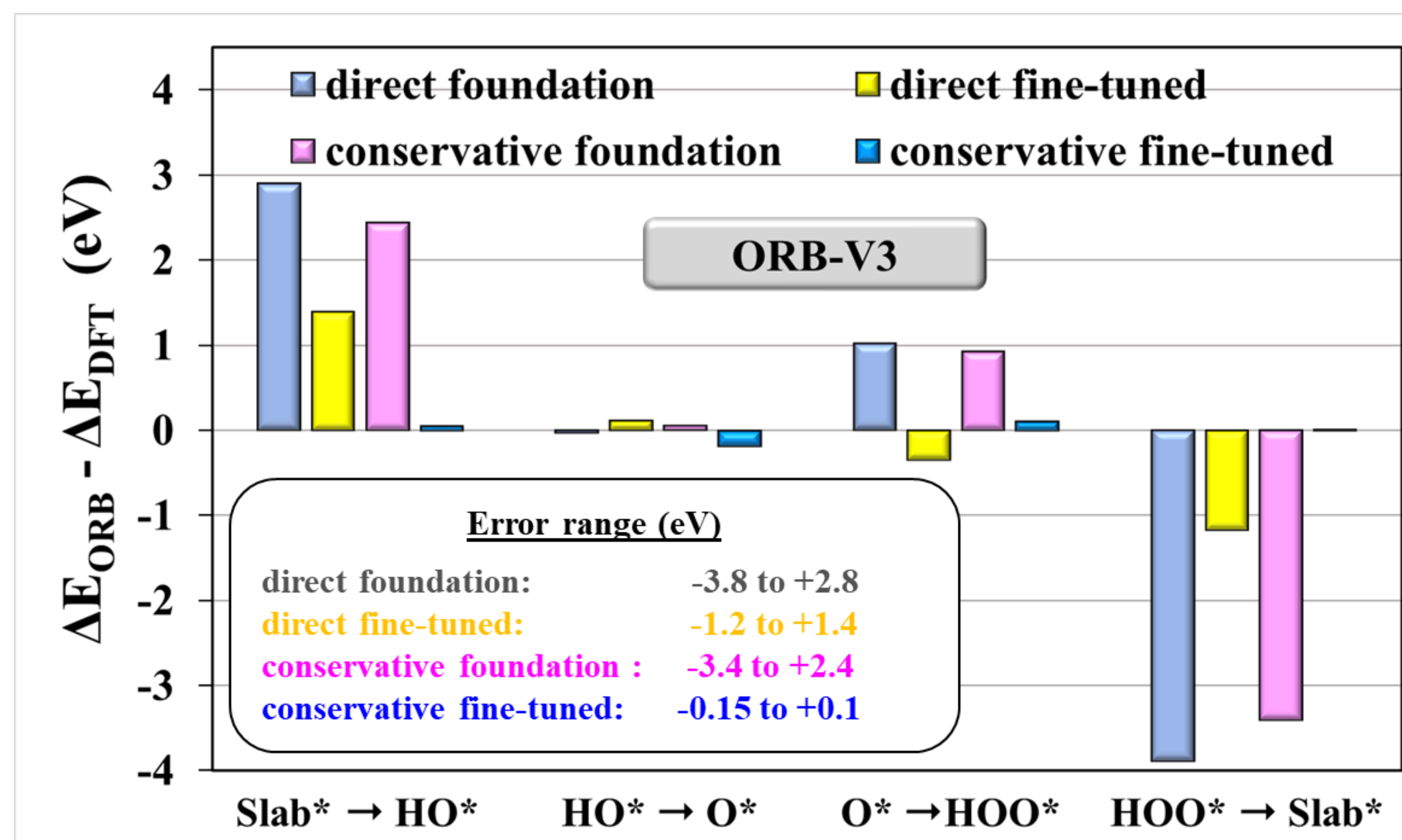


Figure 4. Deviations of ORB-V3 adsorption energies from r²SCAN for the four OER steps on CoOOH/BiVO₄.

- **Foundation ORB-V3** exhibits large deviations from r²SCAN OER energetics.
- **Fine-tuning** dramatically improves adsorption free energies for CoOOH/BiVO₄.
- **Fine-tuned ORB-V3 Conservative** achieves near-quantitative agreement with r²SCAN across all OER steps.
- **Key outcome:** ORB-V3 Conservative outperforms all other models, including MACE, making it the most accurate and transferable MLIP for CoOOH/BiVO₄ OER screening.

Scalability

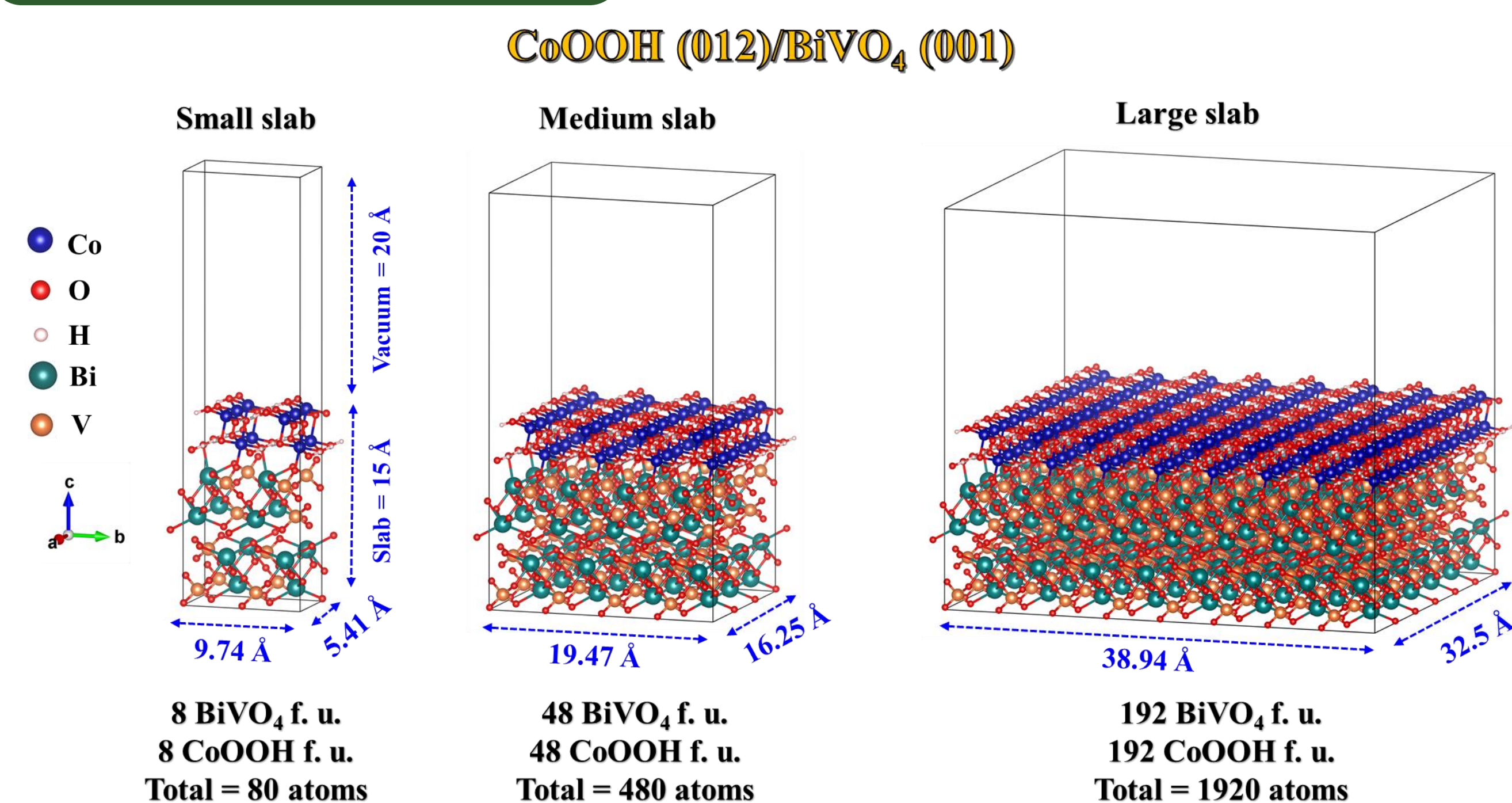


Figure 5. Three CoOOH(012)/BiVO₄(001) supercells: 80, 480 and 1920 atoms.

Size-transferability of ORB-V3 conservative.

Medium- and large-scale slabs are prohibitively expensive for routine r²SCAN (~2,000-8,000 CPU-core-days per geometry), whereas the fine-tuned MLIP reproduces the correct OER energy profile in only minutes.

OER on CoOOH/BiVO₄

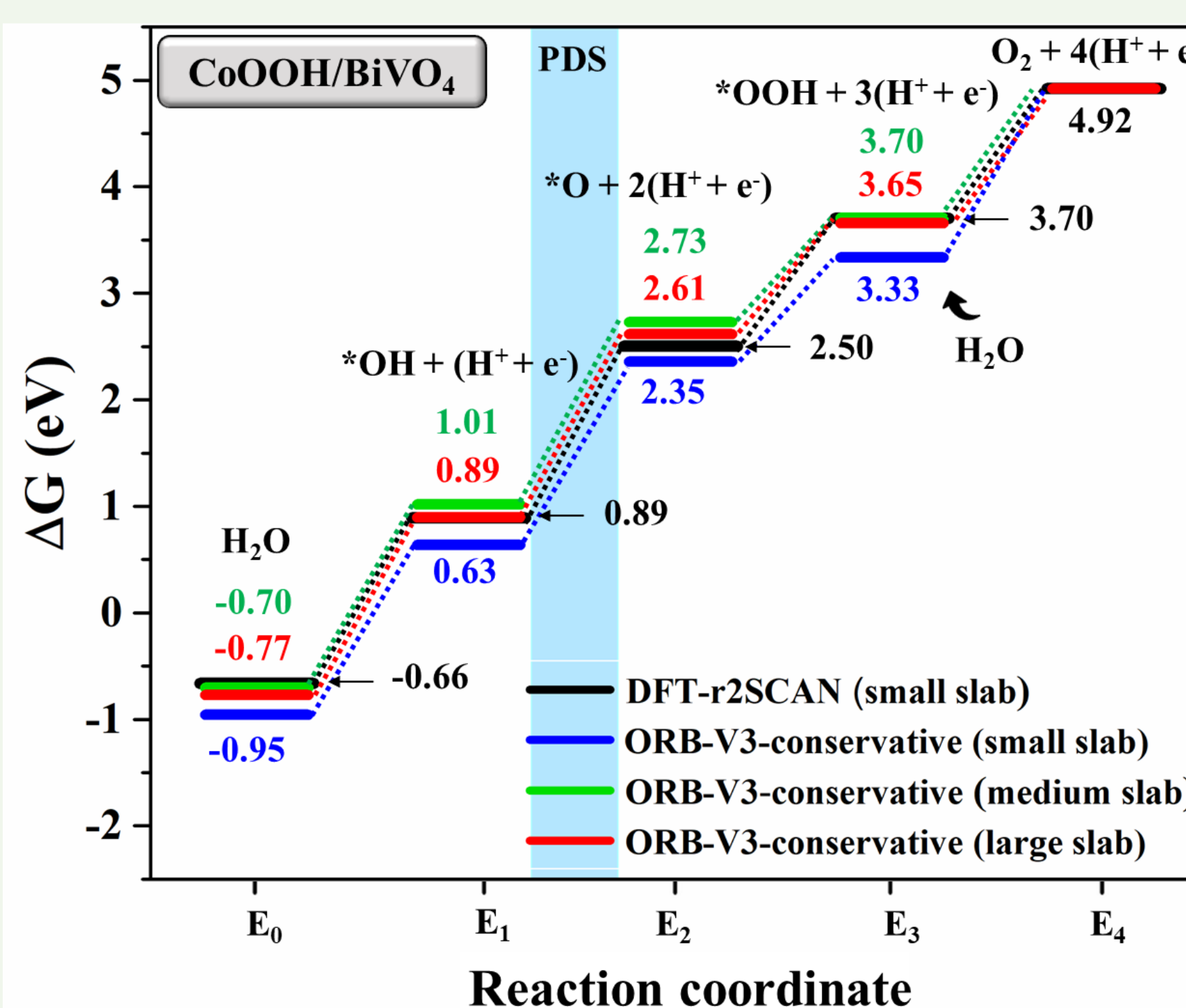


Figure 6. Free-energy diagram on CoOOH/BiVO₄: r²SCAN DFT (small slab) and ORB-V3 conservative (small / medium / large).

➤ Fine-tuned ORB-V3 conservative accurately reproduces the r²SCAN OER free-energy profile across all slab sizes.

➤ The characteristic OER "staircase" profile and potential-determining step are preserved from small to large supercells.

➤ Near-ideal staircase profile: the medium and large slabs differ by only **11 meV** from r²SCAN reference, demonstrating size invariance.

➤ Large-scale OER simulations become feasible with MLIPs, enabling studies beyond the limits of routine DFT.

Conclusion

- Fine-tuned **MACE-MH-1** and **ORB-V3** on a system-specific r²SCAN dataset recover interfacial OER energetics that the foundation models miss by up to 3.8 eV, achieving ~11 meV atom⁻¹ energy MAE and ~0.17 eV Å⁻¹ force MAE on unseen CoOOH/BiVO₄ adsorbate configurations.
- **ORB-V3 conservative** provides the best energy accuracy, force fidelity, and transferability, reproducing the r²SCAN OER free-energy staircase from 80 to 1,920 atoms with sub-0.15 eV step-energy errors while remaining size-invariant.
- **Fine-tuned MLIPs** enable accurate large-scale OER simulations in minutes, extending predictive studies far beyond the practical limits of routine r²SCAN calculations.